

Environmentally Informed GenAI Use in ITSM

A Problem Analysis of Organisational Use-Case Decision-Making

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Abstract: Generative AI (GenAI) is increasingly considered for IT Service Management (ITSM), promising automation, predictive support, service optimisation and improved knowledge processing. At the same time, AI systems create environmental burdens through energy use, emissions, water consumption and hardware-related lifecycle effects. This paper examines environmentally informed GenAI use-case decisions in ITSM. As a problem-analysis echelon within a broader Design Science Research project, it reports a Systematic Literature Review and synthesises literature into six problem characteristics. The study clarifies the focal construct and formulates a literature-grounded problem statement for subsequent objectives and requirements definition.

Keywords: Generative AI (GenAI), IT Service Management (ITSM), digital sustainability, AI governance, problem analysis

1 Introduction

Generative Artificial Intelligence (GenAI) is increasingly adopted across organisations and used to reshape organisational processes, structures and use cases [WEF24]. While GenAI is mainly associated with productivity gains [WEF24], it intensifies a multifaceted governance debate. In particular, existing research highlights the need to consider its consequences, alongside requirements for fairness, transparency and explainability [BI23]. These requirements become more pressing when ecological sustainability is considered. For example, training the underlying deep neural networks is energy-intensive and requires a significant amount of resources [RG25]. Moreover, ecological impacts extend beyond electricity-related emissions to water consumption and impacts across hardware supply chains [RG25], potentially jeopardising sustainability goals [Ga21].

These tensions become particularly actionable in IT Service Management (ITSM), a service-oriented governance domain that aligns business needs with IT capabilities [IE13] and structures decision-making along the IT service lifecycle to improve coordination and service quality [GSF16; Or14]. More recent studies explore deep learning for predictive

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analyses in ITSM, while underscoring prerequisites such as ensuring data quality, process standardisation and system integration [He25]. In contrast, ecological sustainability remains weakly embedded in ITSM research and practice, with only a few contributions addressing this topic [e.g. CT11; SR13]. Overall, related research offers important but fragmented foundations. Green information systems (IS) literature conceptualises IS as socio-technical means for supporting sustainable business practices [Ij10], while digital sustainability broadens this view to the development and deployment of digital resources for environmental, societal and economic welfare [KOS23]. Within this domain, sustainable AI research further highlights energy efficiency, lifecycle management, data management and reduced carbon emissions [Ko25; Le25]. However, these streams do not yet clarify how environmentally relevant evidence should impact concrete organisational decisions on GenAI use cases in ITSM. This creates a problem of translation between environmental evidence and ITSM-specific decision-making.

This paper therefore examines the problem space of environmentally informed GenAI use-case decision-making in ITSM, understood as the organisational process through which responsible stakeholders consider environmentally relevant consequences when deciding on a GenAI-enabled ITSM use case. Accordingly, this study focuses on the following research question: *What characterises the problem of environmentally informed decision-making regarding the use of GenAI in ITSM?* This study is part of a broader DSR project that aims to develop and evaluate decision support for environmentally informed GenAI use-case decisions in ITSM. Following the echeloned DSR logic of [TWV24], the present paper reports the problem-analysis echelon and provides a literature-grounded snapshot of the initial problem understanding and the derived problem characteristics. The aim of this paper is thus to clarify the focal construct, synthesise fragmented literature into problem characteristics and formulate a literature-grounded problem statement. By addressing environmentally responsible decisions on resource-intensive digital technologies, this paper relates to the Sustainable Development Goals (SDGs) [UN15], primarily SDG 12 (Responsible Consumption and Production) and secondarily to SDG 13 (Climate Action).

2 Conceptual background and construct clarification

ITSM is commonly understood as a service-oriented governance domain [IE13] that aligns business requirements with IT capabilities and structures the delivery of IT services in a scalable and quality-assured manner [AMB17]. In this sense, ITSM is associated with improvements in strategy, transparency and service quality across complex organisational settings [GSF16; MT23]. Within this ITSM context, GenAI is increasingly discussed as a transformative technology, e.g. for predictive automation and improved service availability [Si25] or for the usage of AIServiceOps with an emphasis on resilience, proactive insights, customer satisfaction and business wellbeing [VSM23]. Overall, AI in ITSM is mainly discussed in terms of automation, efficiency and service quality – oftentimes in the context of incident management [e.g. VSM23]. The ecological sustainability perspective of including AI into ITSM is only weakly addressed.

In IS research, environmental sustainability is commonly connected to digital sustainability and Green IS. Digital sustainability refers to the development and deployment of digital resources and artefacts toward improving environmental, societal and economic welfare [KOS23], while Green IS focuses on information systems that contribute to sustainable business processes and stresses their socio-technical nature [Ij10]. Although AI for sustainability is already discussed in IS research [Sc23], explicit work on sustainable ITSM or sustainable GenAI use in ITSM remains scarce. Sustainable IT services have been argued to extend beyond economic performance [HA09], ITSM has been discussed as a lever for Green IT [CT11], and practitioners have called for sustainability-oriented extensions of ITSM standards [SR13]. More recent guidance, such as the ITIL sustainability module, acknowledges sustainability in digital and IT contexts but remains largely descriptive [Ax21]. Regarding AI, sustainable AI research highlights energy efficiency, lifecycle management, sustainable data management and reduced carbon emissions [Ko25], while lifecycle-oriented approaches emphasise that sustainable AI cannot be reduced to a single metric or isolated lifecycle phase [Le25]. However, these streams do not yet clarify how environmental evidence should inform concrete organisational decisions on GenAI use cases in ITSM.

Against this background, the focal construct of this study is *environmentally informed GenAI use-case decision-making in ITSM*. It consists of the organisational process through which responsible stakeholders consider environmentally relevant consequences when deciding on the introduction, configuration, continued operation, modification or retirement of a GenAI-enabled ITSM use case. Following the principle of construct clarity [Su10], this construct is deliberately narrower than Sustainable AI, Green AI, AI for Sustainability or Responsible AI. It focuses on the organisational decision problem of relating environmental evidence to ITSM-specific governance responsibilities and decision alternatives rather than on the general sustainability of GenAI systems or AI-enabled sustainability goals. This study addresses this gap by structuring the problem space before deriving solution objectives or a later decision-support artefact.

3 Research Design

Methodologically, this study is positioned within a broader DSR project on Sustainable ITSM. [He04] provide the artefact- and problem-oriented DSR paradigm, while [Pe07] provide the process logic from problem identification to design and evaluation. This paper focuses on reporting a self-contained problem analysis echelon following the **echeloned DSR logic** [TWV24]. The central output is a literature-grounded problem statement that clarifies the characteristics, prior coverage and apparent solvability of environmentally informed GenAI use-case decision-making in ITSM. Accordingly, literature coverage captures which parts of the focal problem are already addressed, whereas remaining problem and solvability capture what remains unresolved and whether the literature indicates plausible structures or mechanisms through which it could subsequently be addressed. The focal construct (section 2) served as the scoping device for the literature

analysis. Publications were assessed according to whether they provided direct or transferable evidence on environmentally relevant organisational decisions concerning GenAI-enabled ITSM use cases. To identify this evidence, a focused Systematic Literature Review (SLR) was conducted following a nine-step protocol based on [KC07]:

First, the **guiding research question** was formulated: “What characterises the problem of environmentally informed decision-making regarding the use of generative AI in IT Service Management?” Second, the **key concepts** were harmonised across the three core components of ITSM, GenAI and ecological sustainability. Broader AI terminology was included to account for heterogeneous wording in the literature, however, the subsequent content-based screening focused on relevance for GenAI-enabled ITSM decision-making. Third, **corresponding keywords** were constructed. Fourth, the **databases** Scopus, IEEE Xplore, ScienceDirect and AIS eLibrary (AISEL) were selected. Fifth, the **search strategy** was defined as title, abstract and keyword searches, restricted to peer-reviewed journal and conference publications. The search string was adapted to database-specific syntax and combined the three components as follows: (*ITSM OR “IT Service Management” OR “incident management”*) AND (*“generative AI” OR “artificial intelligence”*) AND (*sustainab* OR ecologic**). Sixth, **inclusion and exclusion criteria** were specified. The review was limited to English- and German-language publications from 2020 to 2026. Publications were included if at least two of the three core components were central topics and if they contained extractable evidence relevant to the focal construct. Publications were excluded when relevant terms appeared only in venue or conference metadata, when “incident management” referred to non-IT contexts, when sustainability was used without an environmental dimension, or when no extractable evidence for the focal construct was available. These criteria were necessary because the search string intentionally captured directly relevant but also transferable literature. Seventh, the studies were **screened in stages** using a PRISMA-inspired screening flow adapted from [Pa21]. The first stage removed duplicates, records not matching the title or keyword criteria, as well as 12 publications without full-text access. The large reduction of 1621 papers primarily resulted from duplicates and title/keyword screening, particularly because the broad search term “incident management” retrieved numerous non-IT contexts. Subsequent abstract and full-text screening resulted in a total of 22 publications included in the analysis (see Figure 1).

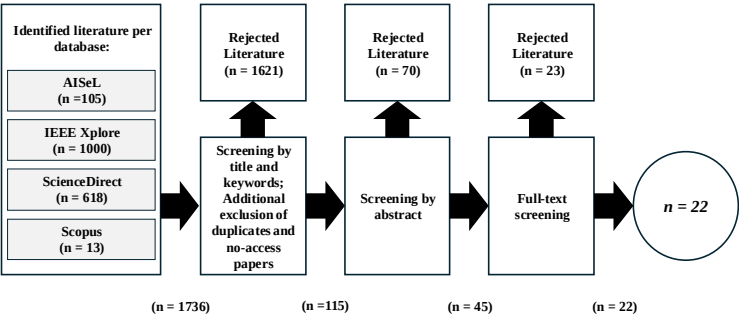


Fig. 1: PRISMA-screening stages based on [Pa21]

Eighth, the 22 publications were **extracted and coded** in a problem-oriented evidence repository covering literature stream, unit of analysis, ITSM relevance, AI scope, environmental scope, problem claims, causes or mechanisms, consequences and remaining gaps. Ninth, the coded evidence was **conceptually synthesised** by comparing recurring problem claims, causes, consequences and remaining gaps across publications and grouping related evidence. These preliminary problem characteristics were iteratively consolidated where they addressed the same decision problem. The resulting problem characteristics were assessed for literature coverage and the apparent solvability of the remaining problem following the problem-analysis echelon logic [TWV24]. The output is therefore a set of literature-grounded problem characteristics and a synthesised problem statement that informs subsequent solution objectives and artefact design requirements.

4 Results of the problem analysis

Overall, the SLR confirms that the intersecting literature of GenAI, ITSM and ecological sustainability is still fragmented rather than absent. Given the narrow focal construct and applied screening criteria, the 22 retained publications provide a focused view of the relevant literature rather than an indication of the field’s overall maturity. They provide direct, transferable or contextual evidence on parts of the problem space, but do not form a coherent research stream on environmentally informed GenAI use-case decisions in ITSM. The inductive synthesis leads to six problem characteristics, assesses their literature coverage and evaluates the apparent solvability of the remaining problem (see Table 1).

Problem characteristic	Literature coverage	Remaining problem and solvability
<i>PC1: Temporal integration deficit</i>	[Be25; Le25; Si25; VSM23]	Environmental evidence is not positioned within ITSM decision points. Moderately solvable through lifecycle structures.
<i>PC2: Lifecycle and boundary fragmentation</i>	[DJ24; Ko25; Le25; LMD21; Ra26; Ro24]	AI impacts are assessed at model or infrastructure level, while ITSM decisions concern use cases. Partly solvable due to boundary and transparency issues.
<i>PC3: Environmental–governance disconnect</i>	[Be25; Ko25; Ma23; MJ24; Yu21; ZT25]	Environmental evidence is not assigned to ITSM roles or authorities. Moderately solvable through role-based governance.
<i>PC4: Ambiguity of direct and indirect effects</i>	[DJ24; Ho24; Os24; Sc23; Vi23]	Direct AI burdens and indirect process effects risk being conflated. Conceptually solvable, empirically difficult.
<i>PC5: Decision operationalisation deficit</i>	[Bo23; GMC24; KB20; Mo25; Si25; VSM23]	Evidence, review points and documentation logic remain unspecified. Moderately solvable through later design work.
<i>PC6: Evidence and capability deficit</i>	[Bo23; Fa24; Ko25; Ma23; Mo25; Ra26; Ro24; ZT25]	Robust indicators, comparable data and interpretive capabilities are insufficiently specified. Partly solvable.

Tab. 1: Problem characteristics (PCs) derived from the SLR

The six problem characteristics reconstruct the focal problem as a set of interrelated translation deficits: environmental evidence needs to be positioned earlier in ITSM decisions (*PC1*), related to explicit lifecycle and system boundaries (*PC2*), connected to ITSM governance responsibilities (*PC3*), differentiated into direct and indirect environmental effects (*PC4*), translated into review points and documentation logic (*PC5*) and supported by comparable evidence and organisational capabilities (*PC6*).

Furthermore, across the reviewed literature, three patterns become visible. AI-enabled ITSM literature explains why organisations adopt AI for automation, service availability, predictive support, resilience, and operational performance, but only weakly connects these decisions to ecological sustainability. Sustainable AI, Green AI, and AI-for-Sustainability literature provide relevant environmental concepts, but often at the level of models, infrastructures, lifecycle phases, or industrial sustainability effects rather than ITSM use-case decisions. Governance- and implementation-oriented literature clarifies adjacent issues such as responsibility, accountability, readiness, and feasibility, but does not yet provide an integrated decision logic for environmentally informed GenAI use in ITSM. The resulting problem statement for this problem analysis echelon can therefore be formulated as follows: *Organisations increasingly consider GenAI-enabled applications for ITSM activities, yet they lack a conceptually and operationally coherent basis for integrating use-case-specific environmental evidence into decisions on introducing, configuring, continuing, modifying or retiring such applications.*

Following the solvability criterion of the problem-analysis echelon [TWV24], this problem is only partially addressed in existing literature but appears moderately solvable because ITSM already provides lifecycle-oriented processes, role-based governance and decision structures into which environmental evidence could potentially be integrated. This indicates plausible starting points for subsequent solution development rather than an already validated solution. However, the reviewed literature does not yet justify the derivation of design requirements or the selection of a specific decision-support artefact.

5 Discussion

The problem analysis suggests that environmentally informed GenAI decision-making in ITSM is not only an unresolved practical challenge, but also reflects a broader conceptual misalignment in the current literature. Environmental AI research often treats impacts as properties of models, infrastructures or lifecycle phases, whereas ITSM research treats decisions as organisational choices embedded in services, roles and governance structures. The central difficulty is therefore not merely to add environmental criteria to ITSM, but to translate evidence produced at one level of analysis into decisions made at another. The relative novelty of organisational GenAI adoption may further explain why this translation problem has only partially been addressed so far, as empirical experience with GenAI-specific ITSM applications is still emerging. However, the review cannot establish this explanation causally and also points to the fragmentation of relevant knowledge. This also

marks the main limitation of the present study. The literature-grounded problem analysis reveals this misalignment and structures its recurring patterns, but it cannot determine how organisations actually prioritise environmental evidence under operational, economic and governance constraints. The apparent solvability of the problem is therefore conditional. ITSM provides promising structures for embedding environmental considerations, but whether these structures can support meaningful decision-making depends on stakeholder responsibilities, data availability and quality, organisational maturity and the willingness to treat ecological sustainability as decision-relevant. The next research step is thus not simply to design a tool, but to validate whether the identified problem characteristics correspond to organisational experience and to determine which form of an artefact can realistically make environmental evidence actionable in GenAI-related ITSM decisions.

6 Conclusion

This paper examined the problem of environmentally informed decision-making regarding GenAI use in ITSM. Based on a SLR, the results show that the problem is not the absence of a relevant knowledge base, but its fragmentation across AI-enabled ITSM, Green IS, Sustainable AI and AI-for-Sustainability research. The analysis identified six interrelated problem characteristics: *temporal integration deficits, lifecycle & boundary fragmentation, an environmental-governance disconnect, ambiguity between direct & indirect environmental effects, decision operationalisation deficits and evidence & capability deficits*. By clarifying the focal construct and formulating a literature-grounded problem statement, the paper contributes a problem-structuring foundation for subsequent DSR, in which the problem statement should be validated with organisational stakeholders before deriving objectives, requirements and an appropriate decision-support artefact.

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